CONSTANTS

$M_e = 5.98 \cdot 10^{24}$ kg

$R_e = 6.37 \cdot 10^{6}$ m

$g = 9.8$ m/s^2

$G = 6.67 \cdot 10^{-11}$ N·m^2/kg^2

$v_{\text{sound}} = 343$ m/s

$m_p = 1.67 \cdot 10^{-27}$ kg

$m_e = 9.11 \cdot 10^{-31}$ kg

$e = 1.60 \cdot 10^{-19}$ C

$k_B = 1.38 \cdot 10^{-23}$ J/K

$R = 8.31$ J/mol-K

$N_A = 6.02 \cdot 10^{23}$ particles/mol

$k = 1/4\pi\epsilon_0 = 8.99 \cdot 10^{9}$ N·m^2/C^2

$\epsilon_0 = 8.85 \cdot 10^{-12}$ C^2/N·m^2

$\mu_0 = 4\pi \cdot 10^{-7}$ T·m/A

$c = 3.00 \cdot 10^{8}$ m/s

$h = 6.63 \cdot 10^{-34}$ J·s

Conversions

$T(°\text{C}) = T(\text{K}) - 273.15$

$1\ \text{m}^3 = 1000\ \text{L}$

1 atm = 101,300 Pa = 760 mm Hg = 14.7 psi

$1\ \text{eV} = 1.60 \cdot 10^{-19}$ J

Basic Formulas

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

KINEMATICS

$\vec{v} = \frac{d\vec{x}}{dt}$ (slope of pos.) $\quad \vec{a} = \frac{d\vec{v}}{dt}$ (slope of vel.)

Circular Motion

$d = \theta R \quad v = \omega R \quad a = \alpha R$

$f = \frac{1}{T} \quad \omega = 2\pi f$

$a_c = v^2/R$

FORCE AND ENERGY

	Force	**Energy**
Gravity (on earth)	$w = mg$	$U_g = mgh$
Gravity (in space)	$F_g = \frac{Gm_1m_2}{r^2}$	$U_g = -\frac{Gm_1m_2}{r}$
Spring	$F_{sp} = k\Delta x$	$U_{sp} = \frac{1}{2}k(\Delta x)^2$
External Force	F_{ext}	$W = F_{\text{ext}}d\cos\theta$
Kinetic Friction	$f_k = \mu_k N$	$W = -f_k d$
Drag	D	
Buoyancy	$F_B = \rho_f V_f g$	
Electric	$F_{\text{el}} = qE$	$U_{\text{el}} = qV$
Magnetic	$F_{\text{mag}} = \lvert q\rvert vB\sin\theta$	$W = 0$
Magnetic	$F_{\text{mag-wire}} = ILB$	$W = 0$
Normal Force	$N \le N_{\text{max}}$	$W = 0$ (for ground)
Tension	$T \le T_{\text{max}}$	$W = Td\cos\theta$
Static Friction	$f_s \le \mu_s N$	$W = 0$ (for ground)
Kinetic Energy		$K = \frac{1}{2}mv^2$
Rotational Kinetic		$K_{\text{rot}} = \frac{1}{2}I\omega^2$
Chemical Energy		E_{chem}
Thermal Energy		E_{th}

Newton's Law

$$\sum \vec{F}_{\text{ext}} = m\vec{a} \qquad \sum \tau_{\text{ext}} = I\alpha$$

ROTATIONAL MOTION

$$x_{\text{cm}} = \frac{x_1 m_1 + x_2 m_2 + \cdots}{m_{\text{tot}}} \qquad y_{\text{cm}} = \frac{y_1 m_1 + y_2 m_2 + \cdots}{m_{\text{tot}}}$$

$$\tau = rF_{\perp} = r_{\perp}F = rF\sin\phi$$

Moments of Inertia

$$I_{\text{pt}} = I_{\text{ring}} = MR^2 \qquad I_{\text{rod ab end}} = \frac{1}{3}ML^2 \qquad I_{\text{rod ab mid}} = \frac{1}{12}ML^2$$

$$I_{\text{disk}} = \frac{1}{2}MR^2 \qquad I_{\text{solid sph}} = \frac{2}{5}MR^2 \qquad I_{\text{sph shell}} = \frac{2}{3}MR^2$$

Rolling

$$v_{\text{cm}} = \omega R$$

$$a_{\text{cm}} = \alpha R$$

MOMENTUM AND ENERGY

$$\vec{p} = m\vec{v}$$

$$L = I\omega$$

$$\vec{J} = \vec{F}_{\text{avg}}\Delta t$$

$$J_{\text{rot}} = \tau_{\text{avg}}\Delta t$$

$$P = \frac{\Delta E}{\Delta t} \text{ or } P = \frac{W}{\Delta t} = Fv$$

Conservation Laws

$$\vec{p}_i + \sum \vec{J}_{\text{ext}} = \vec{p}_f$$

$$L_i + \sum J_{\text{rot ext}} = L_f$$

$$E_i + \sum W_{\text{ext}} + \sum Q_{\text{ext}} = E_f$$

FLUIDS

$$\rho = \frac{m}{V} \qquad P = \frac{F}{A}$$

$$Q = vA \qquad v_1 A_1 = v_2 A_2$$

$$P = P_0 + \rho g h \quad \text{(fluid not moving)}$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$Q = \frac{\pi R^4 \Delta P}{8\eta L}$$

OSCILLATIONS AND WAVES

General Oscillations

$$f = \frac{1}{T} \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

$$x(t) = A\cos(\omega t) \quad v(t) = -\omega A\sin(\omega t) \quad a(t) = -\omega^2 A\cos(\omega t)$$

General Waves

$$f = \frac{1}{T} \qquad \omega = 2\pi f = \frac{2\pi}{T} \qquad k = \frac{2\pi}{\lambda} \qquad v = f\lambda$$

$$y(x,t) = A\cos(kx - \omega t)$$

$$I = \frac{P}{A} = \frac{P_{\text{source}}}{4\pi r^2}$$

Specific Examples

$$\omega_{\text{mass-spring}} = \sqrt{k/m}$$

$$\omega_{\text{pendulum}} = \sqrt{g/L} \qquad \omega_{\text{phys pendulum}} = \sqrt{mgd/I}$$

$$v_{\text{string}} = \sqrt{T/\mu}$$

THERMODYNAMICS

$\rho = \frac{m}{V}$ $P = \frac{F}{A}$

$E_{\text{th-ideal gas}} = \frac{c}{2}Nk_BT = \frac{c}{2}nRT$

$W = -P\Delta V$ (const P) $PV = Nk_BT = nRT$

$PV^\gamma = \text{const}$ (adiabatic) $\gamma = \frac{c+2}{c}$

$K_{\text{avg}} = \frac{3}{2}k_BT = \frac{1}{2}mv_{\text{rms}}^2$ $v_{\text{rms}} = \sqrt{\frac{3k_BT}{m}}$

$\Delta L = \alpha L_i \Delta T$ $\Delta V = \beta V_i \Delta T$

$\epsilon = \frac{W_{\text{out}}}{Q_H} = 1 - \frac{Q_C}{Q_H}$ $\epsilon_{\max} = 1 - \frac{T_C}{T_H}$

Laws

$\Delta E_{\text{th}} = Q + W$ $\Delta S_{\text{univ}} \geq 0$

Heat

$Q = mc\Delta T$ $Q = mL$

$P_{\text{cond}} = \frac{Q}{\Delta t} = \frac{kA}{L}\Delta T$ $P_{\text{rad}} = \frac{Q}{\Delta t} = e\sigma AT^4$

WAVE OPTICS

Light Waves

$I = \frac{1}{2}c\epsilon_0 E_0^2 = \frac{1}{2}\frac{c}{\mu_0}B_0^2$ $v = c/n$ $E_0 = cB_0$

Quantum

$E_{\text{photon}} = hf$

Interference

constructive: extra distance $= m\frac{\lambda}{n}$

destructive: extra distance $= \left(m + \frac{1}{2}\right)\frac{\lambda}{n}$

double slit: extra distance $= d\sin\theta_m \approx d\theta_m$ $\theta_m \approx \frac{y_m}{L}$

thin film: extra distance $= 2t$ const/dest reverse for 1 flip

Diffraction

$w = 2y_1 \approx 2L\theta_1 = \frac{2.44\lambda L}{D}$

GEOMETRIC OPTICS

$n_1 \sin\theta_1 = n_2 \sin\theta_2$ $\theta_c = \sin^{-1}(n_2/n_1)$

$\frac{1}{f} = \frac{1}{s} + \frac{1}{s'}$ $m = \frac{h'}{h} = -\frac{s'}{s}$

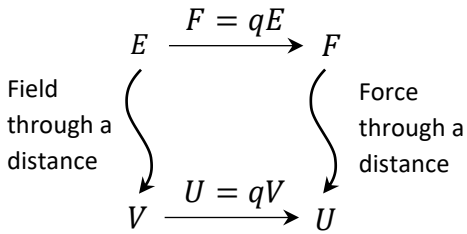
Quantity	Positive when	Negative when
s	Object in front	Object behind
s'	Image where light is (real)	Image not where light is (virtual)
f	Converging lens, concave mirror	Diverging lens, convex mirror
m	Upright	Inverted

Optical Instruments

$M_e = \frac{\theta}{\theta_0} = \frac{25\text{ cm}}{f}$ $m_o = -\frac{L}{f_o}$ $M_{\text{microscope}} = m_o M_e$

$\text{RP} = d_{\min} = \frac{0.62\lambda_0}{\text{NA}}$

FIELDS



$\Delta V = Ed$ (uniform field)

Sources

$E_{\text{pt}} = E_{\text{sph-outside}} = \dfrac{kq}{r^2}$ $\qquad E_{\text{cap}} = \dfrac{Q}{\epsilon_0 A}$

$V_{\text{pt}} = V_{\text{sph-outside}} = \dfrac{kq}{r}$

$B_{\text{wire}} = \dfrac{\mu_0 I}{2\pi r}$ $\qquad B_{\text{loop}} = \dfrac{\mu_0 NI}{2R}$ $\qquad B_{\text{sol}} = \mu_0 I \dfrac{N}{L}$

Induction

$\Phi = BA\cos\theta$ (uniform field)

$\mathcal{E} = -\dfrac{\Delta\Phi}{\Delta t}$

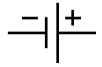
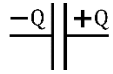
$\mathcal{E} = v\ell B$

CIRCUITS

Kirchhoff's Laws

$\sum I_{\text{in}} = \sum I_{\text{out}}$

$\sum_{\text{loop}} \Delta V = 0$

Component	$\Delta V =$	Sign	Energy
Battery	$\mathcal{E}$		$P = I\mathcal{E}$
Capacitor	Q/C		$U = \dfrac{1}{2}C(\Delta V)^2 = \dfrac{1}{2}\dfrac{Q^2}{C}$
Resistor	IR	ΔV drops going with I	$P = I\Delta V = I^2R = \dfrac{(\Delta V)^2}{R}$

Components

$C_{\text{parallel plate}} = \dfrac{\kappa\epsilon_0 A}{d}$

$R = \dfrac{\rho L}{A}$

Capacitors	Add	Const
Series	$1/C$	Q
Parallel	C	V
Resistors	**Add**	**Const**
Series	R	I
Parallel	$1/R$	V

RC Circuits		$\boldsymbol{\tau = RC}$
Charging	$\Delta V_C = \mathcal{E}\left(1 - e^{-t/\tau}\right)$	$I = I_0 e^{-t/\tau}$
Discharging	$\Delta V_C = (\Delta V_C)_0 e^{-t/\tau}$	